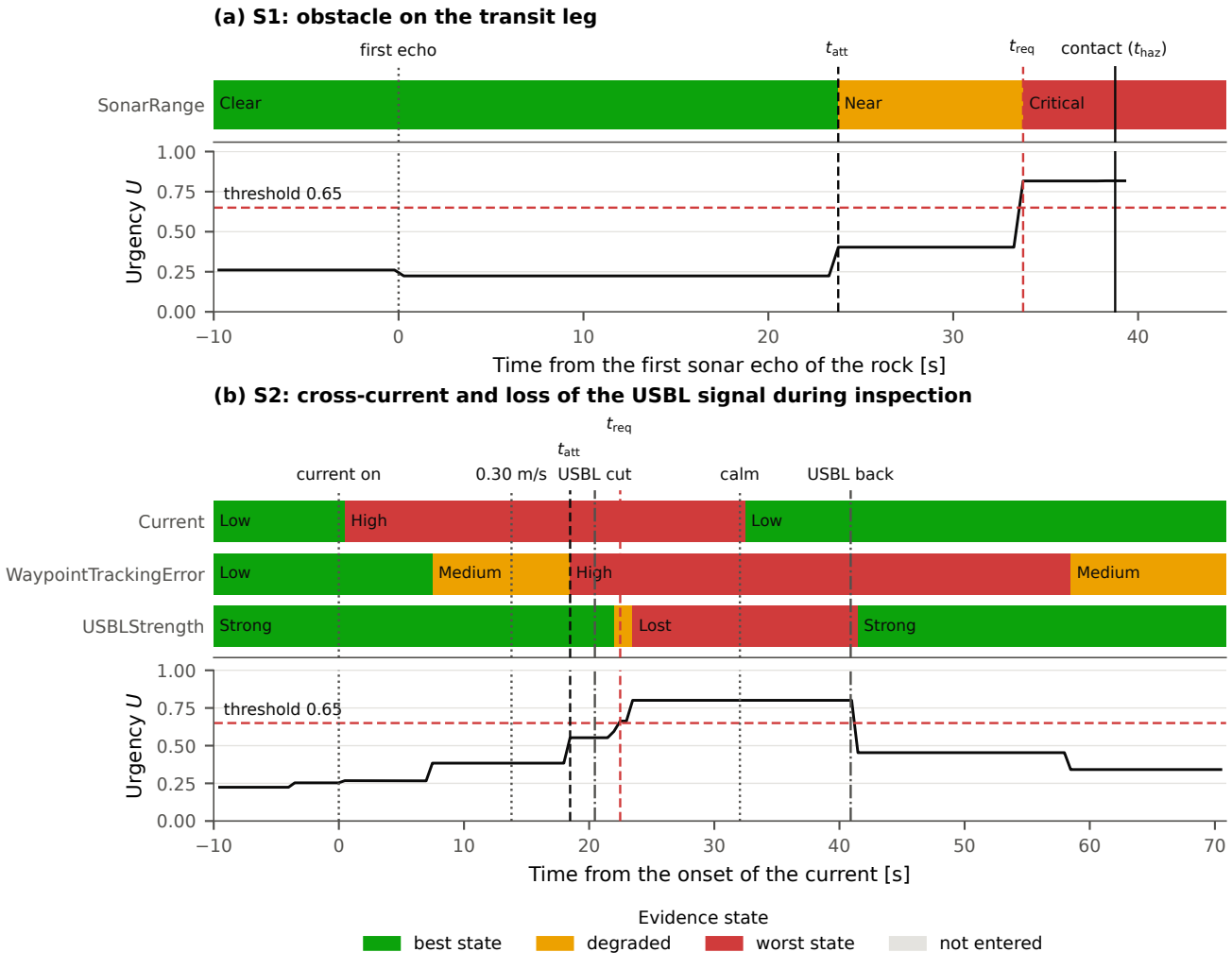
(a) S1: obstacle on the transit leg
first echo
$t_{att}$
$t_{req}$
contact ($t_{haz}$)
SonarRange
Clear
Near
Critical
Urgency $U$
threshold 0.65
1.00
0.75
0.50
0.25
0.00
−10
0
10
20
30
40
Time from the first sonar echo of the rock [s]
(b) S2: cross-current and loss of the USBL signal during inspection
$t_{req}$
$t_{att}$
current on
0.30 m/s
USBL cut
calm
USBL back
Current
Low
High
Low
WaypointTrackingError
Low
Medium
High
Medium
USBLStrength
Strong
Lost
Strong
Urgency $U$
threshold 0.65
1.00
0.75
0.50
0.25
0.00
−10
0
10
20
30
40
50
60
70
Time from the onset of the current [s]
Evidence state
best state
degraded
worst state
not entered